ECO 310: Problem Set 1
Due: Thurs Feb 12

1. You are given the following partial information about a consumer's purchases in two different years (prices are per unit):

   - in year 1, when $p_1 = p_2 = \$10$, she buys 10 units of each good
   - in year 2, when $p_1 = \$10$ and $p_2 = \$7$, she buys 12 units of good 1 and x units of good 2

   For what values of x would you conclude:

   (a) That her year-1 bundle is revealed weakly preferred to the year-2 bundle? What about strictly preferred?

   (b) That her year-2 bundle is revealed weakly preferred to the year bundle? What about strictly preferred?

   (c) That her behavior violates WARP?

2. Suppose Melsi (my puppy) has three toys–F (frisbee), B (ball), and S (stuffed squirrel), and always chooses one when she goes for a walk. When I only put out the frisbee and the ball, she always chooses the frisbee, i.e. $C(\{F,B\}) = \{F\}$.When it's ball or stuffed squirrel, $C(\{B,S\}) = \{B\}$. And when it's frisbee or stuffed squirrel, $C(\{F,S\}) = \{S\}$.

   (a) Explain directly that there's no hope of rationalizing these preferences, i.e. Melsi cannot possibly be making choices to maximize utility. (Hint: if she *were* rational, then saying she will only choose F from $\{F,B\}$, which is what $C(\{F,B\}) = \{F\}$ means, would imply $F \succ B$).

   (b) Prove that regardless of how she chooses from $\{F,B,S\}$ (i.e. no matter what toy she grabs if I put out all three options), she violates IIA.

3. (Quasilinear Preferences) A consumer's preferences over two goods are described by the following utility function:

   $$u(x_1, x_2) = x_2 + \ln x_1$$

   where x_1 is the amount of good 1, and x_2 is the amount of good 2. Her goal is to choose the best affordable bundle, and she has \$2 available. Quantities must be non-negative, but decimals are fine.

   (a) Find the optimal bundle if $p_1 = p_2 = 1$.

   (b) Now find the optimal bundle if $p_1 = 1$ but $p_2 = 3$.

4. Consider a consumer whose utility over leisure time (L) and spending money (S) is given by $u(L,S) = L^{\frac{1}{3}} S^{\frac{4}{3}}$. All spending money is earned by working at wage w per hour. There are T total hours to be divided between work time and leisure time.

   (a) Write out the budget constraint. (How much spending money is available if she wants L hours of leisure?)

   (b) Use the "magic rule for Cobb-Doublas" (will be taught in Lecture 4) to argue that this consumer will choose $\frac{T}{5}$ leisure hours, regardless of the wage.

(c) Explain why these preferences are the same as those represented by $v(L, S) = \log L + 4 \log S$.

(d) Using the utility function in (c), find the optimal demand bundle (L, S) as a function of w and T. Use either a Lagrangian or the intuitive bang-per-buck method.